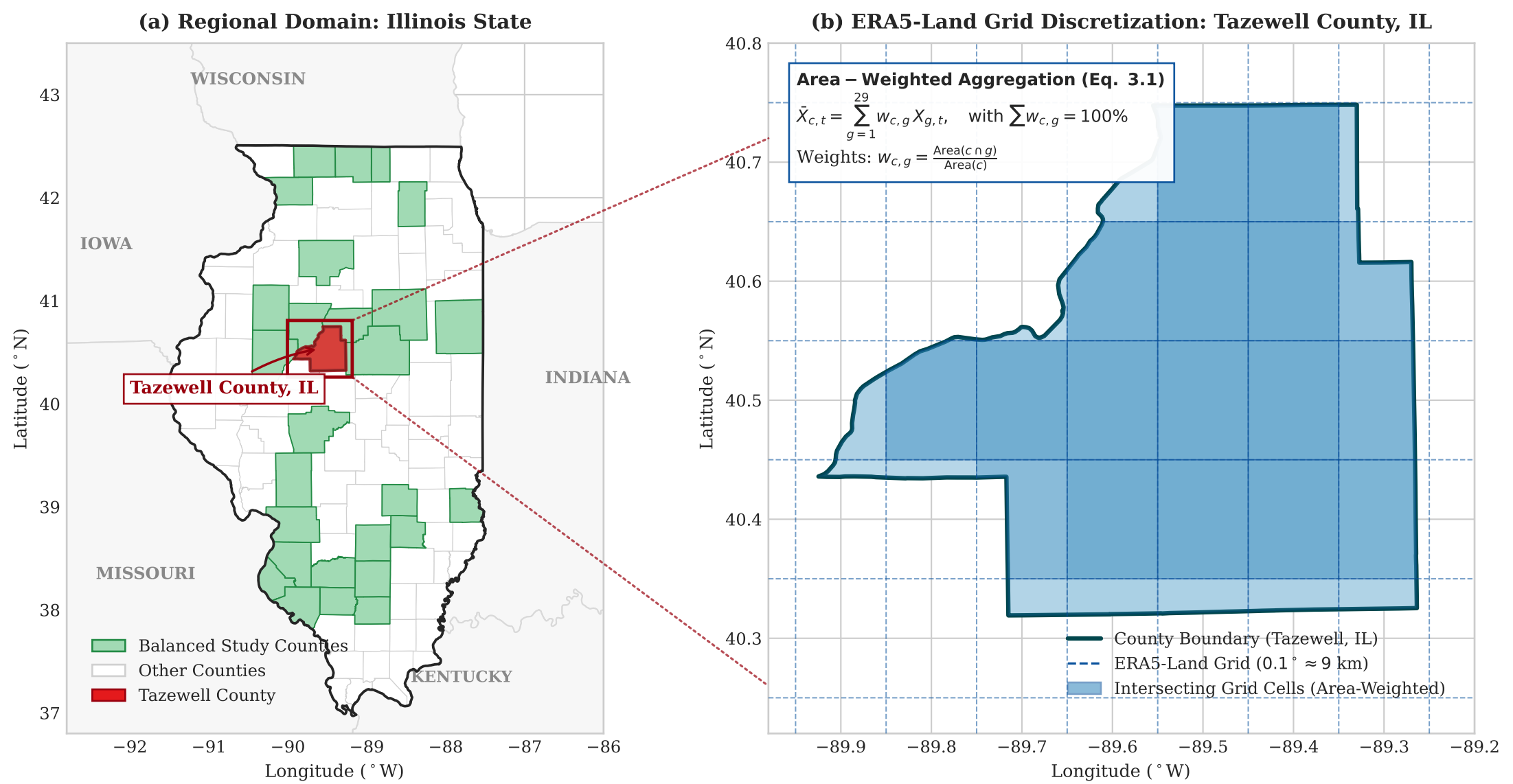

(a) Regional Domain: Illinois State
WISCONSIN
IOWA
INDIANA
MISSOURI
KENTUCKY
Tazewell County, IL
Balanced Study Counties
Other Counties
Tazewell County
Latitude (°N)
Longitude (°W)
(b) ERA5-Land Grid Discretization: Tazewell County, IL
Area – Weighted Aggregation (Eq. 3.1)
$\bar{X}_{c,t} = \sum_{g=1}^{29} w_{c,g} X_{g,t}$, with $\sum w_{c,g} = 100\%$
Weights: $w_{c,g} = \frac{\text{Area}(c \cap g)}{\text{Area}(c)}$
County Boundary (Tazewell, IL)
ERA5-Land Grid (0.1° ≈ 9 km)
Intersecting Grid Cells (Area-Weighted)
Latitude (°N)
Longitude (°W)